National Aeronautics and Space Administration

Goddard Space Flight Center

CUBESAT APPLICATION FOR PLANETARY ENTRY MISSIONS (CAPE)

Jaime Esper[1], Jean-Pierre Baumann[2], Georg Herdrich[2]

[1]NASA Goddard Space Flight Center, Greenbelt MD 20771; Email: Jaime.Esper@nasa.gov

[2]Institute of Space Systems*, Raumfahrtzentrum Baden-Württemberg, University of Stuttgart, Pfaffenwaldring 29, 70569 Stuttgart, Germany

* Institut für Raumfahrtsysteme, IRS

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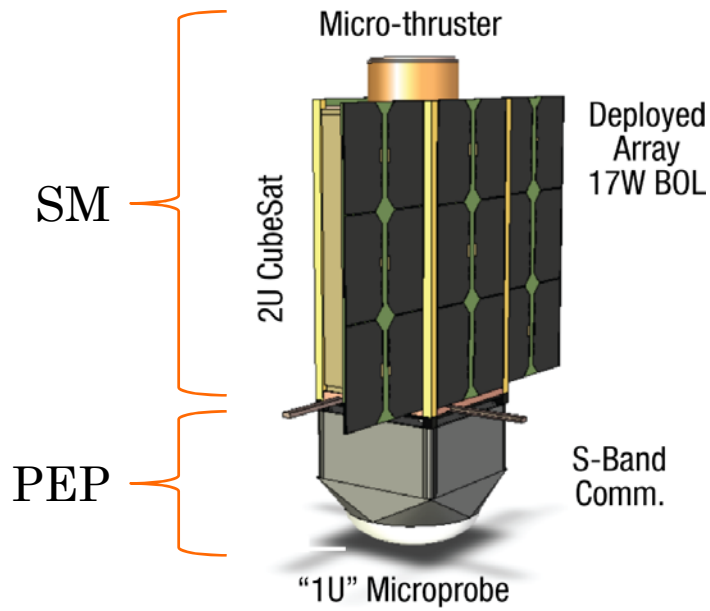
INTRODUCTION

- The Cubesat Application for Planetary Entry Missions (CAPE) concept describes a high-performing Cubesat system
  - includes a propulsion module and miniaturized technologies capable of surviving atmospheric entry heating, while reliably transmitting scientific and engineering data.
- The Micro Return Capsule 2 (MIRKA2) is CAPE's first planetary entry probe flight prototype
- This presentation summarizes CAPE's configuration and typical operational scenario.
- It also summarizes MIRKA2's design and basic aerodynamic characteristics

CAPE Configuration

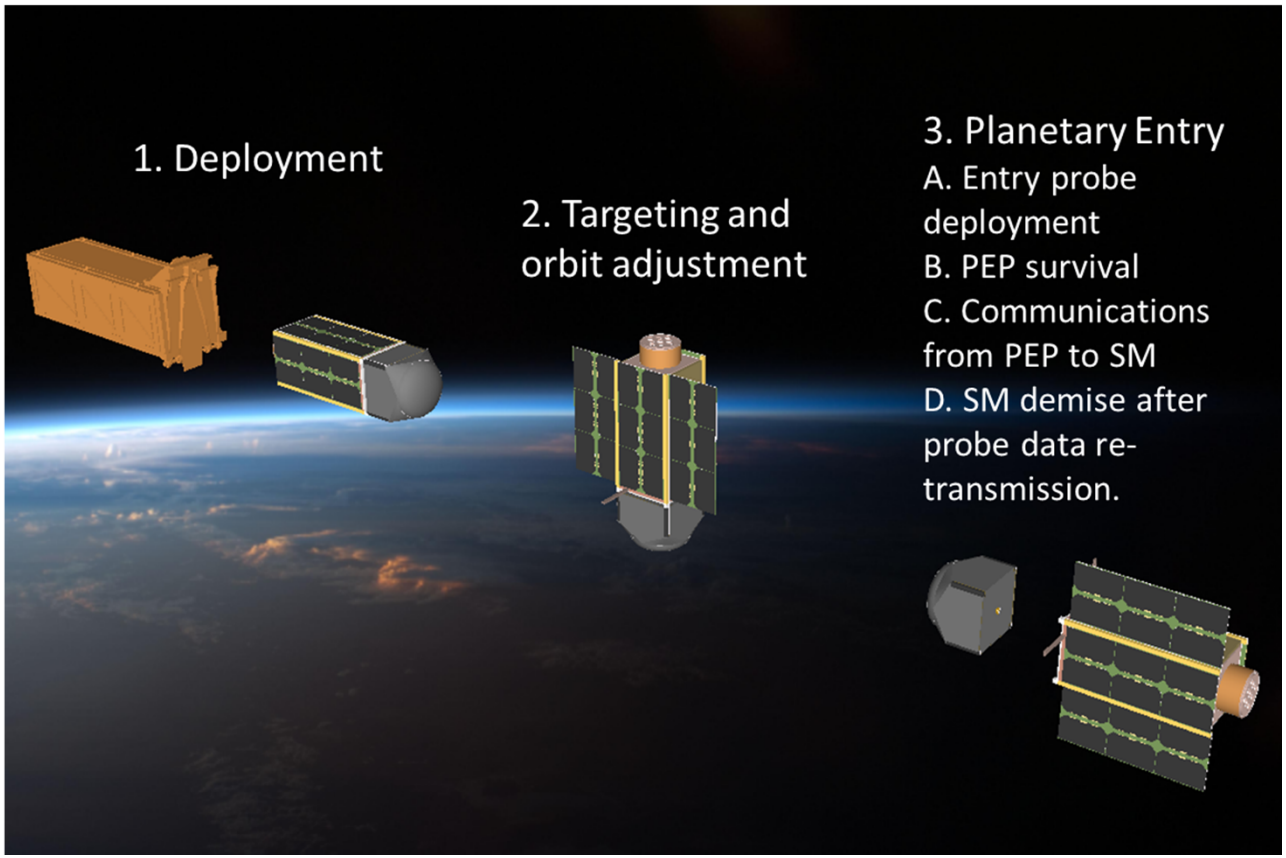
- Service Module (SM): contains the subsystems necessary to support vehicle targeting (propulsion, ACS, computer, power) and the communications relay unit.
- Planetary Entry Probe (PEP): carries the scientific instrumentation and Thermal Protection System (TPS) technology monitoring.

The total system mass is less than 5 kg. The solar array generates about 17W at 1 AU, and the system nominally consumes about 11W of power.



Operations Concept

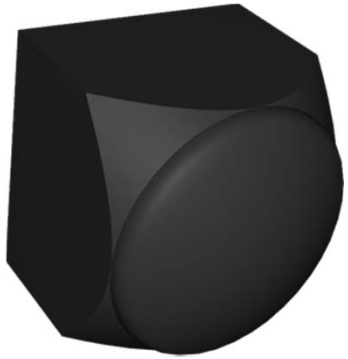
- Three mission phases are identified: 1. Deployment, 2. Targeting, and 3. Planetary Entry.



MIRKA2

- Micro Return Capsule (German acronym is MIRKA2, for Mikro-Rückkehrkapsel).
- First PEP re-entry demonstrator for CAPE.
- Currently being designed by NASA GSFC and the IRS in Germany.
- Expected to validate key system technologies, including advanced miniaturized sensors and a new thermal protection system (TPS) material developed by GSFC in collaboration with the IRS.

MIRKA2 Aeroshell design with Resin Impregnated Carbon Ablator (RICA) cover



CAPE-MIRKA2 (CM2)

- Mission Objective: use the Low Earth Orbit (LEO) environment to demonstrate key operational aspects involved in de-orbit and atmospheric entry of a miniature planetary probe.
- University participation exemplifies the potential in bringing students and organizations with limited resources within the realm of planetary exploration.
- For analysis purposes, the following mission assumptions are made:
  - Launch in September 2014, at 40º inclination.
  - Propulsive deorbiting with an on/off duty cycle of 50% over an 85 minute orbit.
  - Δv at separation of 5 m/s.

MIRKA2 Design Concept

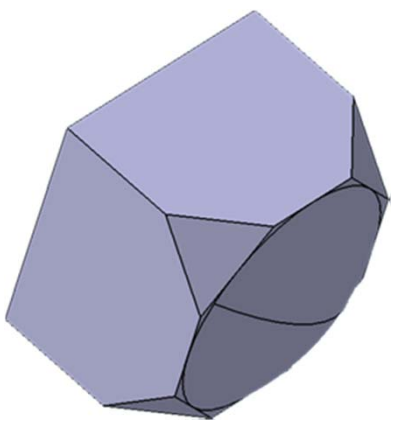
- The aeroshell design is driven by the Cubesat form-factor constraints.
- The subsystem arrangement is defined by the requirement to situate the CG as much forward of the CP as possible to ensure a stable flight.
- The TPS heat shield consists of three different layers
  - Outer layer 5 mm thick made of the RICA
  - Variable thickness low-density insulator (DLR)
  - 3mm-thick aluminum "cold" structure.
- Sensors included are
  - Analog Resistance Ablation Detector (ARAD) sensors
  - Radiometer
  - Thermocouples
  - Pressure Transducer
  - Flux (Φ) Probe Experiment (FIPEX)

MIRKA2 AERODYNAMICS

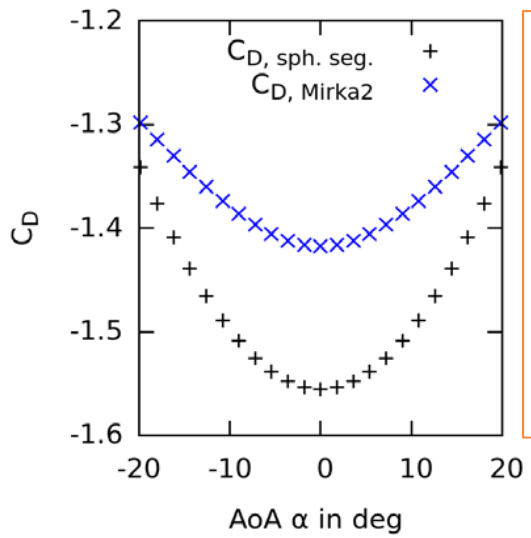
- Flight stability is one of the most important aspects in MIRKA2's performance since it affects heating, communications, and in the end, mission success.
- To estimate stability and attitude, angular momentum equations are applied over the vehicle body during flight.
- Torques used in these equations were taken from pitching moment coefficients of "comparable" missions and environments.
- To assess MIRKA2's aerodynamic behavior, the continuum, transitional and free-molecular flow regimes are all considered in a way similar to the approach used for Stardust and MUSES-C.

Hypersonic Continuum Regime and Drag Coefficient (C_D) Results

- To be able to obtain an analytical solution for the C_D, the aeroshell surface is simplified and modeled as a sphere segment and plane surfaces.



MIRKA2 Aeroshell Approximation



Comparison of MIRKA2 and a hemispherical segment drag coefficient as a function of angle of attack (90 km altitude).

The C_D values for MIRKA2 are similar to an equivalent spherical segment, but notably slightly *smaller* at Angles of Attack (AoA) between ±20°

Transitional and Free-Molecular Flow Regimes - Discussion

- Analytical calculations for the transitional and free-molecular flow regimes are still on-going.
- The following items must be considered during the course of the analysis:
  - Aerodynamic analyses of other entry vehicles revealed significant instabilities might remain down to the continuum, resulting in severe heating of less protected areas exposed to the flow-field.
  - Because of its size, transitional properties can then persist longer and to lower altitudes for MIRKA2, which means that critical flight instabilities can potentially persist longer as well.
  - In addition to vehicle stability in a somewhat idealized flow-field, there are environmental uncertainties to consider, such as cross-wind, atmospheric density, initial state vector, vertical wind, axial force coefficient in the hypersonic continuum (C_A), and Δv at separation.

Summary and Outlook

- CAPE enables a broad new class of planetary entry and in situ missions, and brings planetary exploration capability to resource limited organizations, including universities and small businesses.
- CM2 will demonstrate the operational scenario involved in deploying a micro-probe in a planetary environment.
- MIRKA2 will serve to demonstrate the first Cubesat-sized entry vehicle.